CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 5: **Exploring Forces** | Quick revision, one page

Force

- A force is a push or a pull; SI unit: newton (N).
- A force can change speed, direction of motion or shape of an object.
- Net force on two forces in the same direction $= F_1 + F_2$; in opposite directions $= F_1 - F_2$ (larger minus smaller).
- Balanced forces: net force is zero; unbalanced forces change the state of motion.

Types of forces

- Contact: muscular, friction. Non-contact: gravitational, magnetic, electrostatic.
- Electrostatic force: between charged bodies; like charges repel, unlike attract; charged bodies also attract neutral ones.

Friction

- Opposes the relative motion of surfaces in contact; acts opposite to the motion.
- Kinds: static, sliding, rolling; rolling friction $<$ sliding friction.
- Increased by rough surfaces and a greater push; reduced by lubricants, polishing, wheels and ball bearings.
- Fluid friction (drag): acts on objects moving through liquids and gases.

Mass and weight

- Weight $W = mg$; near the Earth, about 10 N per kg is used in this chapter.
- Mass: kg (constant). Weight: N (changes with gravity).
- Weight is measured by a spring balance.

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